# The Coding Exam

Short Report

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## 1 Part A — Base Model

### 1.1 Optimal solution

The model was coded following equations (1)–(10) exactly. One thing I did differently for constraint (6): instead of adding an explicit row, I passed `ub=200` when declaring the `inventory` variables.

Gurobi solved the instance to optimality in under 0.01 s.

**Optimal objective: $7 029.84**

Table 1 shows where the money goes. Production and furnace activation together take up almost 90% of the total, which makes sense—the $200 activation fee per furnace-period adds up fast when you're running two furnaces. The back-order cost ended up at zero, not because it was infeasible to delay deliveries, but because pre-producing and holding stock turned out cheaper than paying the $4–5 per-unit penalty.

Table 1: Cost breakdown — base model

| Component | Cost ($) | Share (%) |
|---|---|---|
| Production | 4 939.28 | 70.3 |
| Transportation | 535.00 | 7.6 |
| Activation(Furnace) | 1 200.00 | 17.1 |
| Overtime | 146.79 | 2.1 |
| Holding | 208.77 | 3.0 |
| Back-orders | 0.00 | 0.0 |
| **Total** | **7 029.84** | 100.0 |

Tables 2 and 3 show the full production plan and furnace usage.

Table 2: Optimal plan: $D_{k,t}$, $X_{k,t}$, $Y_{k,t}$, $I_{k,t}$, $B_{k,t}$

| $k$ | $t$ | $D_{k,t}$ | $X_{k,t}$ | $Y_{k,t}$ | $I_{k,t}$ | $B_{k,t}$ |
|---|---|---|---|---|---|---|
| A | 1 | 55 | 82.53 | 55.00 | 27.53 | 0.00 |
| A | 2 | 57 | 61.90 | 57.00 | 32.43 | 0.00 |
| A | 3 | 61 | 46.43 | 61.00 | 17.86 | 0.00 |
| A | 4 | 63 | 45.14 | 63.00 | 0.00 | 0.00 |
| B | 1 | 36 | 63.27 | 36.00 | 27.27 | 0.00 |
| B | 2 | 39 | 47.45 | 39.00 | 35.72 | 0.00 |
| B | 3 | 52 | 35.59 | 52.00 | 19.31 | 0.00 |
| B | 4 | 46 | 26.69 | 46.00 | 0.00 | 0.00 |
| C | 1 | 24 | 36.11 | 24.00 | 12.11 | 0.00 |
| C | 2 | 36 | 38.87 | 36.00 | 14.98 | 0.00 |
| C | 3 | 31 | 29.15 | 31.00 | 13.14 | 0.00 |
| C | 4 | 35 | 21.86 | 35.00 | 0.00 | 0.00 |

Looking at Table 2, the model over-produces in periods 1 and 2 for all three products, building up inventory that gets consumed as demand rises in periods 3–4. By period 4, inventory is exactly zero—which is just the terminal condition $B_{k,4} = 0$ doing its job; there's no point holding stock beyond the last period.

Table 3: Furnace schedule

| Furnace | $t$ | $z_{m,t}$ | Reg. hrs | OT hrs |
|---|---|---|---|---|
| F1 | 1 | 1 | 50.00 | 0.00 |
| F1 | 2 | 1 | 50.00 | 0.00 |
| F1 | 3 | 1 | 50.00 | 9.79 |
| F1 | 4 | 1 | 50.00 | 0.00 |
| F2 | 1 | 1 | 50.00 | 0.00 |
| F2 | 2 | 1 | 29.71 | 0.00 |
| F2 | 3 | 0 | — | — |
| F2 | 4 | 0 | — | — |

F1 runs at exactly 50 regular hours every period. The only overtime in the entire solution is 9.79 hrs on F1 in period 3, right at the demand peak. F2 maxes out in period 1, uses only 29.71 hrs in period 2, and then shuts down.

## 1.2 Sensitivity analyses

### Holding cost ($\alpha$ sweep)

I swept $\alpha \in [0, 5]$ over 20 points, replacing $CH_k$ with $\alpha \cdot CH_k$. Figure 1 plots total cost and total inventory against $\alpha$.

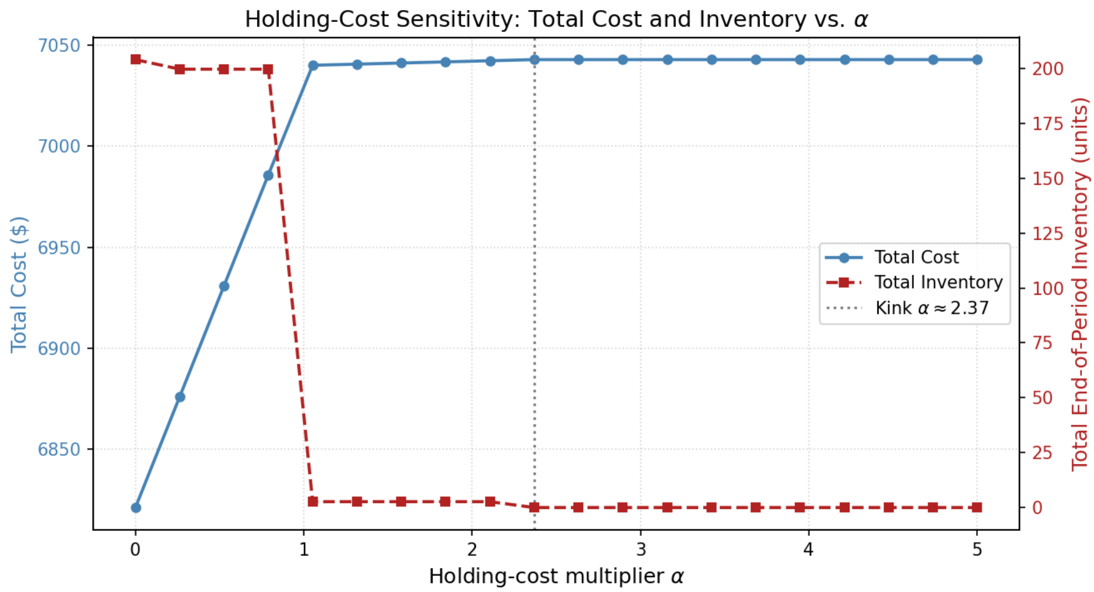


Figure 1: Total cost and end-of-period inventory vs. $\alpha$.

Up to $\alpha \approx 1.053$, the model holds around 200 units and cost climbs steeply—unsurprisingly, since the same stock pile gets pricier to store. At $\alpha \approx 1.053$ inventory falls to $\sim$2.67 units in a single step, and at $\alpha \approx 2.37$ it reaches zero. Past that point the total cost freezes at \$7 042.92 no matter how large $\alpha$ gets.

### Overtime cost ($\beta$ sweep)

I swept $\beta \in [0.5, 4.0]$ over 15 points, replacing $c^{ov}_{m,t}$ with $\beta \cdot c^{ov}_{m,t}$. Figure 2 shows the result.

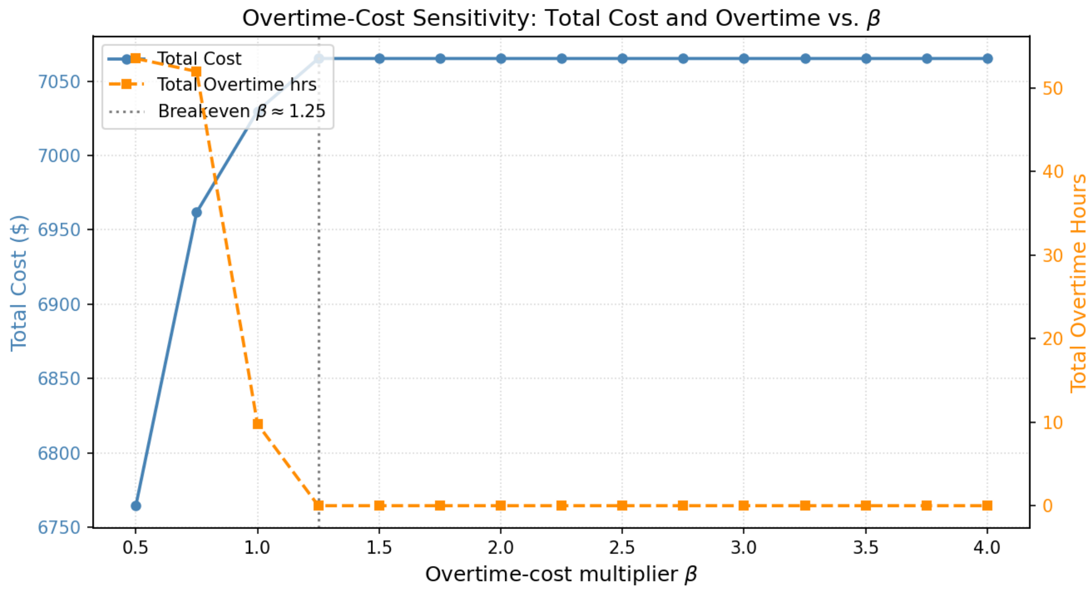


Figure 2: Total cost and overtime hours vs. $\beta$.

Overtime goes to zero at $\beta \approx 1.25$, which corresponds to \$18.8/hr. After that the cost stabilises at \$7 065.18. That's meaning that the overtime premium is high enought, so the model prefers to pay the activation fee and regular hours.

### 1.3 Discussion

**Why does the holding-cost curve kink?** not answered.

**Is F2 used in all periods?** No, only in periods 1 and 2. Keeping it off in periods 3 and 4 saves \$400 in activation fees. It's worth it because by period 3 the smoothing constraint caps how much can be produced anyway (at most $1.25 \cdot X_{k,2}$), so the extra F2 capacity would mostly go unused. F1 with under 10 hrs of overtime is enough, and F2's higher unit cost ($c_0^{F2} = 10$ vs. $c_0^{F1} = 8$) makes it a bad deal unless capacity is really tight.

**What do the smoothing constraints actually do?** Without them, the cheapest plan would be to dump everything into one period on F1, pay one activation fee, and let the furnaces sit idle for the rest of the horizon. The $\pm 25\%$ bounds prevent that: once production drops, it can only recover slowly, which forces the model to keep furnaces running through the whole planning period. From an operations standpoint that's exactly right—you can't spin a steel furnace back up overnight.

## 2 Part B — Minimum Lot Size and Setup Cost

### 2.1 Mathematical formulation

**New binary variables.**

$$\delta_{m,k,t} \in \{0, 1\}, \quad \forall m \in M,\ k \in K,\ t \in T.$$

$\delta_{m,k,t} = 1$ means furnace $m$ produces product $k$ in period $t$, paying a setup; zero means it doesn't touch that product that period.

**Coupling constraints.**

$$x_{m,k,t} \le M_{m,k,t}\, \delta_{m,k,t} \quad \forall m \in M, k \in K, t \in T \tag{B1}$$

$$x_{m,k,t} \ge L_k\, \delta_{m,k,t} \quad \forall m \in M, k \in K, t \in T \tag{B2}$$

Constraint (B1) forces $x_{m,k,t} = 0$ whenever $\delta_{m,k,t} = 0$ (Big-$M$ side). Constraint (B2) forces $x_{m,k,t} \ge L_k$ whenever $\delta_{m,k,t} = 1$ (minimum-lot side)

**Modified objective.**

$$\min\ (\text{eq. }(1))\ +\ \sum_{m,k,t} SC_{m,k}\,\delta_{m,k,t}, \qquad SC_{m,k} = 50\ \forall\,(m,k).$$

The eq. (1) corresponds to the original objective.

### 2.2 Results and comparison

Table 4: Part A vs. Part B — summary

| | Part A | Part B |
|---|---|---|
| Optimal cost ($) | 7 029.84 | 7 637.39 |
| Cost increase ($) | — | +607.56 |
| Setups paid | — | 12 times |

F1 sets up for A and C every period (8 setups), adds B in periods 3 and 4 (2 more), and F2 runs B only in periods 1–2 (2 setups): 12 total. Nothing else was worth activating given the $50 fee plus the minimum-lot commitment.

Table 5: Aggregate production $X_{k,t} = \sum_m x_{m,k,t}$: Part A vs. Part B

| $k$ | $t$ | $X_{k,t}^{(A)}$ | $X_{k,t}^{(B)}$ | $\Delta$ |
|---|---|---|---|---|
| A | 1 | 82.53 | 74.09 | −8.44 * |
| A | 2 | 61.90 | 67.14 | +5.24 * |
| A | 3 | 46.43 | 50.35 | +3.93 * |
| A | 4 | 45.14 | 44.42 | −0.73 * |
| B | 1 | 63.27 | 63.27 | 0.00 |
| B | 2 | 47.45 | 47.45 | 0.00 |
| B | 3 | 35.59 | 35.59 | 0.00 |
| B | 4 | 26.69 | 26.69 | 0.00 |
| C | 1 | 36.11 | 32.39 | −3.73 * |
| C | 2 | 38.87 | 40.48 | +1.61 * |
| C | 3 | 29.15 | 30.36 | +1.21 * |
| C | 4 | 21.86 | 22.77 | +0.91 * |

* changed from Part A.

Product B doesn't change at all because neither furnace was below $L_B = 20$ in Part A. Products A and C shift around because small sub-threshold lots get dropped: the F2 lot for A in period 1 was $x_{F2,A,1} = 11.42 < L_A = 25$, so it disappears and the missing volume gets redistributed to F1 over the remaining periods—within the smoothing bounds.

### 2.3 Discussion

**Why does Part B cost more?** 12 setups at $50 accounts for $600 of the $607.56 gap. The remaining $7.56 comes from the plan being slightly worse overall: some lots that were small but efficient in Part A now either have to be dropped or inflated to meet $L_k$, and since the smoothing constraints link all four periods together, those adjustments ripple forward and make the whole schedule a bit less efficient. It's a small effect here, but with stricter lot sizes it would matter more.

**Does a very large $SC_{m,k}$ necessarily reduce setups?** Not in this case. The F2-B setups in periods 1 and 2 aren't there because they were cheap—they're there because F1 is already running 50 hours on A and C and has no capacity left for B.